12.3 Mathematical Logic

George Boole was a self-taught English Mathematician, Philosopher and Logician. His results on **Boolean Algebra** involving the binary numbers play an important role in various fields, particularly more in computer applications. He introduced the idea of Symbolic Logic and contributed a lot of results to the fast development of Mathematical Logic.



George Boole
(1815-1864)

The reputed Greek philosopher Aristotle (384-322BC(BCE)) wrote the first book on logic. The famous German philosopher and mathematician Gottfried Leibnitz of 17thcentury framed the idea of using symbols in Logic. Later this idea was realized by George Boole and Augustus de Morgan in 19th century. George Boole established the fact that logic is very much related to mathematics by linking logic, symbols, and algebra together. Mathematical Logic was developed in the late 19thand early 20thcenturies.

In 1930 the researchers noticed (Neumann's statement in his death bed: *0 and 1 are going to rule the world*) that the binary numbers 0 and 1 could be used to analyze electrical circuits and thus used to design electronic computers. Today digital computers and electronic circuits are designed to implement this binary arithmetic. We study Mathematical Logic as the language and deductive system of Mathematics and Computer Science.

Generally Logic is the study of valid reasoning. But mathematical logic allows us to represent knowledge in a precise mathematical way and it also allows us to make valid inferences using a set of precise rules. It is regarded as a powerful tool for computer science because it is mainly used to verify the correctness of programs.

12.3.1 Statement and its truth value

The simplest part of Mathematical Logic is the **Propositional Logic** and its building blocks are statements or propositions. Mostly communication needs the use of language through which we impart our ideas. They are in the form of sentences.

There are various types of sentences like

(1) Declarative (Assertive type)

(2) Imperative (A command or a request type)

(3) Exclamatory (Emotions, excitement type)

(4) Interrogative (Question type)

(5) Open type

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Definition 12.7

Any **declarative sentence** is called a **statement** or a **proposition** which is either **true** or **false** but not both.

Any **imperative sentence** such as exclamatory, command and any **interrogative sentence** cannot be a proposition.

The **truth value** of a statement refers to the truth or the falsity of that particular statement. The truth value of a true statement is **true** and it is denoted by T or 1**.** The truth value of a false statement is **false** and it is denoted by F or 0**.**

An **open sentence** is a sentence whose truth can vary according to some conditions, which are not stated in the sentence. For instance, (i) $x \times 7 = 35$ is an open sentence whose truth value depends on value of x. That is, if $x = 5$, it is true and if $x \neq 5$, it is false. (ii) *He is a bad person*. This is an open sentence. Opinion varies from individual to individual.

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